Nazia Tasnim

Ph.D. Student, Computer Science, Boston University
Expected graduation: Fall 2027 | **Seeking internships and full-time roles**
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Research Interests

I study how visual and multimodal models represent the world, where those representations fall short, and how to adapt them efficiently. My research connects **spatial-reasoning evaluation** and **representation interpretability** with **parameter-efficient adaptation and compression**: evaluating what models understand, examining the representations behind their behavior, and developing resource-efficient ways to adapt them. Current work extends this agenda to **audio imagination for spatial navigation**. Complementary research in *adversarial robustness*, *multilingual safety evaluation*, and *machine unlearning* examines the gap between observed behavior and underlying vulnerabilities or retained knowledge.

Technical Skills

- **Languages:** Python, SQL, Bash, PySpark, Java, JavaScript, SPARQL.
- **Frameworks & Libraries:** PyTorch, Hugging Face (Transformers, PEFT), OpenCV.
- **Inference & Systems:** vLLM, RAG Pipelines, Git, Docker.
- **Methods & Focus:** Post-Training & Fine-Tuning (SFT, LoRA, DPO, PEFT), Multimodal Learning, Latent Probing (Patchscopes, LatentLens), Safety & Alignment.

Selected Publications

1. **Seeing Isn't Orienting: A Cognitively Grounded Benchmark Reveals Systematic Orientation Failures in MLLMs** | ECCV'26
   Nazia Tasnim*, Keanu Nichols, et al.
   - Developed an object-orientation benchmark with **33K+** questions across **7** tasks, revealing systematic spatial-understanding failures in **22** VLMs.
2. **Decompose, Mix, Adapt: A Unified Framework for Parameter-Efficient Neural Network Recombination and Compression** | CVPR'26
   Nazia Tasnim*, Shrimai Prabhumoye & Bryan A. Plummer
   - Unified PEFT and compression through shared-basis recombination, improving over SOTA PEFT by **1.5%** with **1.3K** trainable parameters per task and compression baselines by **5.5%** at **50%** compression.
3. **RECAST: Reparameterized, Compact Weight Adaptation for Sequential Tasks** | ICLR'25
   Nazia Tasnim* & Bryan A. Plummer
   - Developed a reparameterization method using fewer than **50** task-specific trainable parameters, outperforming SOTA by up to **3%**.
4. **VISTA: Vision Transformer Enhanced by U-Net and Image Colorfulness Frame Filtration for Automatic Retail Checkout** | CVPR Workshop'22
   Nazia Tasnim*, Istiak Shihab, et al.

- Built a real-time retail product-recognition pipeline achieving **46%** accuracy and ranking **3rd** on the cross-modality dataset.

5. **Vision-LLMs Can Fool Themselves with Self-Generated Typographic Attacks** | `NeurIPS Workshop'24`
   Maan Qraitem, **Nazia Tasnim**, et al.
   - Introduced **two** self-generated typographic attacks that reduce vision-language classification performance by up to **33%**.
6. **Representation-Aware Unlearning via Activation Signatures: From Suppression to Knowledge-Signature Erasure** | `Findings of EMNLP'26`
   Syed Naveed Mahmood, ..., **Nazia Tasnim**, et al.
   - Introduced **KIF** for representation-aware unlearning, achieving near-oracle erasure while preserving utility; evaluated behavioral suppression versus knowledge removal across **3B–14B** models.
7. **From Facts to Folklore: Evaluating Large Language Models on Bengali Cultural Knowledge** | `ACL-IJCNLP'26`
   Nafis Chowdhury, ..., **Nazia Tasnim**, et al.
   - Developed **BLanCK**, covering **5,265** terms across **16** categories, revealing gaps in Bengali cultural knowledge and gains from contextual information.

Professional Experiences

IVC-ML Group, Boston University **Sept'23 – Present**
Graduate Researcher

- Investigating **audio imagination for spatial navigation**.
- Developed **DORI** to evaluate object-orientation understanding in MLLMs (*ECCV'26*).
- Developed a unified framework for **parameter-efficient adaptation and compression** (*CVPR'26*).
- Developed **RECAST** for compact weight adaptation across sequential tasks (*ICLR'25*).
- Developed self-generated typographic attacks to assess **VLM robustness** (*NeurIPS Workshop'24*).

Adobe Inc., San Jose, CA **Summer'26**
AI/ML Intern, Customer Experience Orchestration

- Developed pipelines for processing **structured and free-form insights** generated by Adobe Coworker.
- Designed an interpretable, controllable **CJA insight reranking and diversification** protocol for low-latency processing.

Wikimedia Foundation **Sept'22 – June'23**
Natural Language Processing Data Scientist

- Developed **resource-efficient NLP tools** supporting **300+ languages** across 100+ wiki projects.
- Built pipelines to curate community resources and produce reusable datasets, models, and tools for internal research.

Ongoing Research and Submissions

Visual Representation Decoding in VLMs | *Ongoing research collaboration*

- Developing a unified evaluation framework for **visual-representation decoding**, comparing readout methods to identify complementary failures in object and attribute recovery.

Component-wise Rank-1 Refinement (CR1R) | *Under review*

- Contributing to **training-free visual refinement** for charts and tables, outperforming ViCrop by up to **3.60 points** across ten benchmarks.

Multilingual Medical-Safety Evaluation | *Under review*

- Investigating **multilingual medical-safety failures** across prompt framings and medical topics, assessing whether direct-query refusals generalize to reframed requests.

Selected Projects

mwtokenizer [Package]
A multilingual **Python tokenization package** for Wikimedia Projects focusing on non-whitespace delimited languages. Analyzed large-scale wikicorpora through **PySpark** to optimize pattern recognition in diverse writing systems and curated essential module assets.

mwparserfromhtml [Package]
A **Python** library to parse and extract metadata from Enterprise HTML Dumps. The module is part of the core components of the **Mediawiki Utilities Project**.

VISTA [Code]
Developed an end-to-end pipeline for real-time retail checkout, combining **UNET**-based segmentation with *entropy masking* for domain bias reduction. Implemented multi-class product classification using **Vision Transformers (ViT)**. Designed a **custom metric** for frame selection, optimizing object detection efficiency in video streams.

Shaako [Code] [Demo]
A mobile application to provide rural people with Emergency Medical Care support through community health workers. Built the frontend in **Android Framework**. The backend portal is **Django** and uses both **Firebase** and **Azure CMS** for database.

Honors and Awards

- Recipient of **WiCV Mentorship Grant** | CVPR'25 & '26
- Recipient of **CRA GradCohort Fellowship** | Cohort 2025
- Recipient of **Dean's Fellowship**, Boston University GSAS. | Fall'2023 - Summer'2024
- **Outreachy Internship** at the Wikimedia Foundation (top 1.4% applicant) | Summer'22
- Second Runner-up at **Microsoft Imagine Cup** - SEANM | 2022
- Second Runner-up, **CVPR AI City Challenge** Track-4 (3rd globally) | 2022
- **Grace Hopper Scholar** | 2021

Leadership and Community Service

- **Research Mentorship:** Mentored **10+** undergraduate and master's students leading to publication, including as Research Instructor for the **NSF REU Summer Program (2025)**.
- **Instruction & Teaching:** Teaching Fellow at **Boston University** for graduate courses in NLP, AI, and Cybersecurity.
- **Conference Reviewing: ICML, IJCV, ICLR, ECCV, CVPR, ACL-ARR**.